Name: _____ Period: _____

Part 1 · Vectors at Any Angle

Optional practice for students who have shown the Target tier. These problems come from the honors course and go past what the general unit tests: vectors at any angle, aiming across a current, launches from a height, velocity at impact, and derivations in symbols. Each part says which sections to finish first. Use Desmos for any time you cannot solve directly.

Tier: Extension **When:** After general Sections 2.3 and 2.4

Learning targets

- Add vectors that point in any direction, not just north, south, east, and west.
- Explain with symbols why components work.

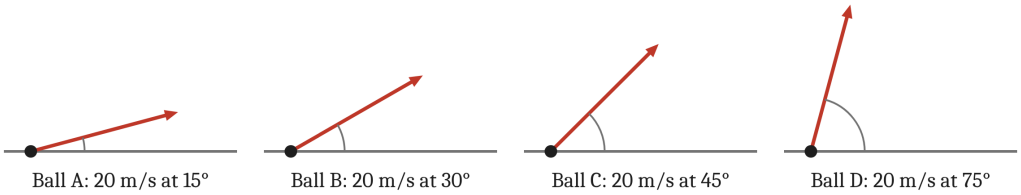
Key ideas

- Break every vector into x and y parts with $A_x = A\cos\theta$ and $A_y = A\sin\theta$, using a sketch for the signs. Add the x parts, add the y parts, then find R with the Pythagorean theorem and its direction with $\tan^{-1}$.
- Two vectors can add to anything from the difference of their sizes (opposite directions) to the sum of their sizes (same direction).

1. Challenge. Mrs. Kraus jumps off the ground at 40° above the horizontal with a vertical velocity component of 3.0 m/s. Find her launch speed and her horizontal velocity component.

2. Symbolic. A vector of magnitude A points at angle θ above the +x axis. Write A_x and A_y in terms of A and θ. Then show that $A_x^2 + A_y^2 = A^2$. Which right-triangle fact did you just use?

3. Ranking. Mrs. Luksic hits four softballs, each at 20 m/s, at the angles shown. Rank the balls, greatest first.



Ball A: 20 m/s at 15° Ball B: 20 m/s at 30° Ball C: 20 m/s at 45° Ball D: 20 m/s at 75°

a) v_{ix}: Greatest _____ _____ _____ _____ Least ☐ All equal

b) v_{iy}: Greatest _____ _____ _____ _____ Least ☐ All equal

c) Which two balls have components that swap places? _____ and _____ Explain why.

4. $A = 30$ m east and $B = 20$ m at 60° N of E. Fill in the table and find $A + B$.

Vector	x component	y component
A		
B		
R		

5. A hiker walks 4.0 km at 30° N of E, then 6.0 km at 45° W of N. Find the hiker's resultant change in position.

6. What single walk would take the hiker in problem 5 straight back to the start?

7. Add 8.0 m at 0° and 5.0 m at 120° (measured counterclockwise from +x).

8. Add three vectors: 10 m north, 15 m at 40° S of E, and 8.0 m west.

9. A rescue boat travels 3.0 km at 30° N of E, then 5.0 km at 20° W of N. A helicopter at the boat's starting point flies straight to the boat at 120 km/h. Find the helicopter's heading and the flight time in minutes.

10. Challenge. $A + B = 10$ m north, and $A = 6.0$ m at 30° N of E. Find B (magnitude and direction).

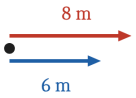
11. Symbolic. Dr. Shafer and a partner drag a heavy sled with two ropes. Each pulls with the same force F (a vector, in newtons), one rope at angle θ to the left of straight ahead and one at θ to the right.

a) Find the forward and sideways components of each pull. b) Show that the combined pull is $2F\cos\theta$ straight ahead.

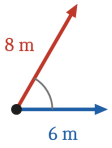
c) What is the combined pull when $\theta = 0°$? When $\theta = 90°$? Why should the two pullers keep their ropes close together?

12. Ranking. In each case a 6 m vector and an 8 m vector are added. Only the angle between them changes. Rank the cases by the magnitude of the resultant, greatest first, without calculating.

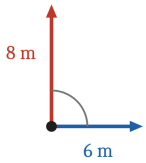
Case A: angle 0°



Case B: angle 60°



Case C: angle 90°



Case D: angle 180°



Resultant: Greatest _____ _____ _____ _____ Least ☐ All equal

Check case C with the Pythagorean theorem: R = _____ Explain the pattern.

13. Can a 5.0 m vector and an 8.0 m vector add to a resultant of a) 2.0 m, b) 13 m, c) 10 m? Explain each with a sketch.

Part 2 · Relative Velocity at an Angle

Tier: Extension **When:** After general Section 2.5

Learning targets

- Find where a boat lands, and how to aim it to land directly across.
- Solve crosswind problems for a plane's heading and ground speed.

Key ideas

- Velocity relative to the ground = velocity relative to the water (or air) + velocity of the water (or air).
- To land directly across, aim upstream so the upstream part of the boat's velocity cancels the current: $v_b \sin\theta = v_c$. The part left over, $\sqrt{v_b^2 - v_c^2}$, carries the boat across.

1. Rain falls straight down at 8.0 m/s. You drive at 15 m/s. Relative to you, how fast is the rain moving, and at what angle from vertical do its streaks appear on the side window?

2. Symbolic. A boat with speed v_b relative to the water points straight across a river of width w. The current is v_c. Write the crossing time, the downstream drift, and the angle of the path from straight across. Which of the three change if the current doubles?

3. Qualitative. Does a stronger current help, hurt, or not affect a swimmer who points straight across? Answer separately for the crossing time and the landing point, and explain.

4. A boat moves at 5.0 m/s relative to the water. The current is 3.0 m/s, and the river is 80 m wide.

a) At what angle upstream must it aim to land directly across?

b) What is its speed relative to the bank? c) How long does the crossing take?

5. A boat (4.0 m/s relative to the water) aims 20° upstream on a 60 m river with a 2.0 m/s current. How long does it take to cross, and where does it land?

6. A pilot wants to travel due north. The plane's airspeed is 250 m/s and the wind blows toward the east at 50 m/s. Find the heading and the ground speed. How long does a 500 km trip take?

7. A plane heads 30° N of E at 180 m/s through a wind blowing toward the south at 40 m/s. Find its velocity relative to the ground.

8. A swimmer who swims at 1.0 m/s tries to cross a river whose current is 1.5 m/s. Can she land directly across? Explain using vectors.

9. Symbolic. A boat with speed v_b relative to the water must land directly across a river of width w with current $v_c (v_c < v_b)$. Show that the aim angle upstream is $\theta = \sin^{-1}\left(\frac{v_c}{v_b}\right)$ and the crossing time is $t = \frac{w}{\sqrt{v_b^2 - v_c^2}}$. What happens to t as v_c gets close to v_b? Why does that make sense?

10. Ranking. Four planes, each with an airspeed of 200 m/s, must travel due north. Each flies in a different wind that blows toward the east. Rank the planes, greatest first.

Plane	Airspeed	Wind (toward the east)
A	200 m/s	0 m/s
B	200 m/s	20 m/s
C	200 m/s	40 m/s
D	200 m/s	60 m/s

a) Angle the plane must turn west of north: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Ground speed: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain both rankings.

11. Rain streaks on a car's side window make a 60° angle with the vertical when the car moves at 20 m/s. How fast is the rain falling?

12. Challenge. A pilot must fly to a city 400 km away at 30° N of E. The plane's airspeed is 220 km/h, and a 40 km/h wind blows toward the south. Find the heading the pilot must use and the time the trip takes.

Part 3 · Horizontal Launches: Going Further

Tier: Extension **When:** After general Sections 2.6 to 2.9

Learning targets

- Find the velocity (size and direction) of a projectile when it lands.
- Derive landing distances in symbols and use a graph to measure g.

Key ideas

- Velocity at impact: v_x stays v_{ix}; find v_{fy} from $v_{fy}^2 = v_{iy}^2 + 2a_y(y_f - y_i)$. Then $v = \sqrt{v_x^2 + v_{fy}^2}$ and the angle below horizontal is $\tan^{-1}(|v_{fy}| \div v_x)$.
- A **Symbolic.** problem asks for an answer in letters. Work the same x and y columns, but keep h, v, and g as letters.

1. Challenge. In the lab, a group forgets the cup's height. Their table is 0.90 m high, the cup is 0.12 m tall, and $v_{ix} = 2.4\ \text{m/s}$. Where do they place the cup, and where should it go? Does the ball fall short or go long?

2. Symbolic. A ball rolls off a table of height h at speed v. Show that it lands a distance $x = v\sqrt{(2h\ /\ g)}$ from the table. By what factor does x change if h is made four times larger? If v is doubled?

3. Ranking. Four balls roll off tables. Rank them, greatest first, and explain each ranking with the x world or the y world.

Ball	Table height h	Launch speed v
A	1.0 m	2.0 m/s
B	1.0 m	4.0 m/s
C	4.0 m	1.0 m/s
D	0.25 m	4.0 m/s

a) Fall time: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Landing distance: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain.

4. Symbolic. A plane flying level at speed v and height H drops a package. Derive how far before the target it must release the package. Then show that the line of sight from the plane down to the target at release makes an angle ϕ below horizontal with $\tan\phi = \dfrac{\sqrt{\frac{gH}{2}}}{v}$.

5. Qualitative. A ball rolls off a table on the Moon ($g = 1.6\ \text{m/s}^2$) at the same speed and from the same height as on Earth. By what factor do the fall time and the landing distance change? Explain without plugging in numbers for h or v.

6. Challenge. Marble A rolls off a 0.80 m table at 3.0 m/s. Marble B rolls off a different table at 2.0 m/s and lands the same distance from its table. How tall is B's table?

7. Challenge. Mr. Lehman rides a motorcycle off the end of a ramp, moving horizontally 5.0 m above the ground. The landing platform is 2.0 m tall and its near edge is 12 m away horizontally. What is the minimum launch speed?

8. Mr. Lehman runs horizontally off a 20 m cliff at 2.5 m/s and dives into the water. Find how far out from the cliff Mr. Lehman lands and the velocity (magnitude and direction) entering the water.

9. A ball is launched horizontally at 6.0 m/s. At what time is its velocity directed 45° below the horizontal?

10. Symbolic. A ball is launched horizontally at speed v from height h. Show that it hits the ground at speed $v_f = \sqrt{v^2 + 2gh}$, at an angle below horizontal with $\tan\theta = \dfrac{\sqrt{2gh}}{v}$.

11. Challenge. A stone thrown horizontally from a 20 m cliff hits the water at 25 m/s. Find its launch speed.

12. Linearizing data (class lab graph). Five groups rolled marbles off the 0.85 m lab table at different launch speeds and measured the landing distance x_f from the point below the table edge.

a) Plot x_f (vertical axis) against launch speed v_{ix} (horizontal axis) on the grid. Draw one straight best-fit line.

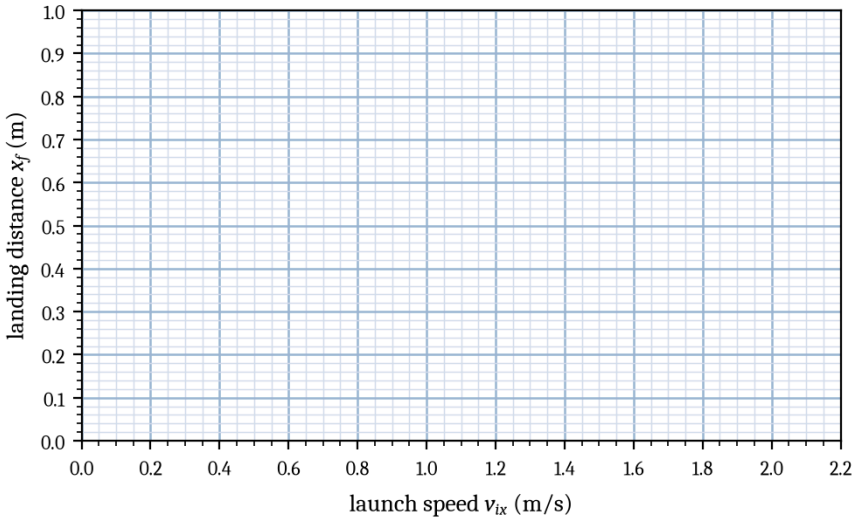
b) Find the slope of your line, using two points on the line (not data points). Include units.

c) Use the x column, $x_f = v_{ix}t + x_i$ with $x_i = 0$, to explain why the slope is the fall time t.

d) Use the calculated time from the slope of your x_f vs. v_{ix} graph to calculate g and find the percent difference from 9.8 m/s².

e) Why should the best-fit line pass through the origin?

Trial	1	2	3	4	5
Launch speed v_{ix} (m/s)	0.82	1.09	1.41	1.68	1.97
Landing distance x_f (m)	0.35	0.44	0.59	0.69	0.83



Part 4 · Angled Launches from a Height

Tier: Extension **When:** After general Sections 2.10 and 2.11

Learning targets

- Solve angled launches that start or end above the ground.
- Use the area under av_y graph to find heights.

Key ideas

- Put the ground at $y = 0$. A launch from a height h has $y_i = h$, and a landing on the ground has $y_f = 0$.
- The y equation $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ now has at^2 and a t term. Graph $y = -4.9x^2 + v_{iy}x + y_i$ in Desmos and click where the curve reaches the landing height. A curve that crosses a height twice gives two times: going up and coming down.
- Gravity pulls everything down the same $\frac{1}{2}gt^2$ below where it would be without gravity.

1. At the zoo, Mr. Peck films a kangaroo whose jump reaches a maximum height of 2.5 m. If it jumps at 45°, what is its launch speed and how far does it travel horizontally?

2. Ranking. Four balls are launched on level ground. Fill in v_{iy} for each first, then rank them, greatest first.

Ball	Launch speed	Angle	v_{iy}
A	20 m/s	30°	
B	20 m/s	60°	
C	10 m/s	90° (straight up)	
D	14.1 m/s	45°	

a) Time in the air: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Maximum height: Greatest _____ _____ _____ _____ Least ☐ All equal

c) Range: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain which launch value sets each ranking.

3. A cannon on a 25 m cliff fires at 40 m/s, 30° above the horizontal, toward a boat at sea level.

a) Find the maximum height above the water. b) Find the time to reach the water.

c) How far from the base of the cliff is the boat? d) Find the velocity at impact.



4. For the cannon in problem 3, explain why the time to reach the water is more than twice the time to reach the top.

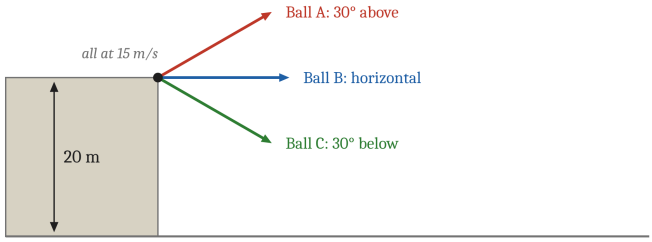
5. Angry Birds. A bird is launched from ground level at 18 m/s at 50°. A 6.0 m tall wall stands 25 m away. How high is the bird when it reaches the wall? Does it clear the wall?

6. Mrs. Luksic throws a softball at 15 m/s at 60°. It lands on a roof 6.0 m above the launch point while coming down. Find the time in the air, the horizontal distance, and the velocity at landing.

7. A stone is thrown from a 30 m bridge at 12 m/s, 20° **below** the horizontal. Find when and where it hits the water and its speed at impact.

8. Mr. White puts a shot at 13 m/s from 2.1 m above the ground. Find the horizontal distance at 40° and at 45°. Which angle is better, and why isn't 45° automatically best here?

9. Ranking. Dr. Shafer throws three medicine balls from the top of a 20 m cliff, each at 15 m/s, in the directions shown. Rank them, greatest first.



a) Time in the air: Greatest _____ _____ _____ Least ☐ All equal

b) Speed at impact: Greatest _____ _____ _____ Least ☐ All equal

Check one impact speed with components. Does the result surprise you? Explain.

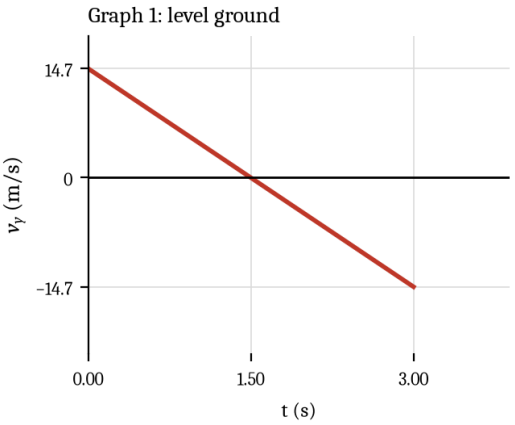
10. Area under a velocity–time graph (angled launch). Graph 1 is v_y vs. t for a ball launched at an angle with $v_{iy} = 14.7$ m/s that lands at its launch height. Graph 2 is the same launch from the top of a cliff; the ball lands at the base.

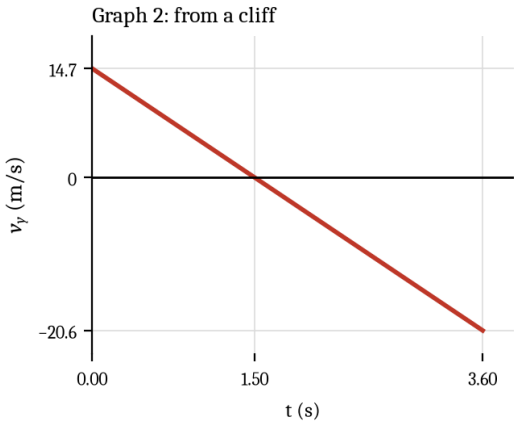
a) On graph 1, shade the area that equals the maximum height above the launch point, and compute it.

b) Find the total area on graph 1 (area above the axis minus area below). Explain why it is zero on level ground.

c) On graph 2, find the area above the axis and the area below the axis. Use the net area to find the cliff height.

d) Check your cliff height with the y column: $y_f = \frac{1}{2}a_y t^2 + v_{iy} t + y_i$.





11. Monkey and hunter (concept). A dart gun is aimed straight at a target hanging from a branch. The target is released at the instant the dart is fired. Ignoring air resistance, does the dart pass above the target, hit it, or pass below it? Explain by comparing each object with where it would be without gravity.

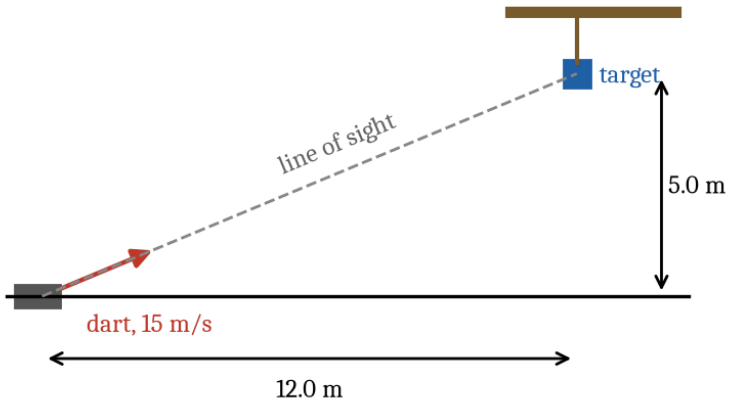
12. Monkey and hunter (numbers). The target hangs 12.0 m away horizontally and 5.0 m above the dart's launch height. The dart is aimed straight at it and leaves at 15 m/s; the target drops at the same instant. Use x and y columns.

a) Find the launch angle, v_{ix}, and v_{iy}. b) How long does the dart take to reach the target's x position?

c) At that time, how far below the line of sight is the dart? How far has the target fallen?

d) Where do they meet (height above the launch point)?

e) What is the slowest launch speed that still hits the target before it falls to the launch height?



Part 5 · Working Backward and Deriving

Tier: Extension **When:** After general Section 2.11

Learning targets

- Work backward from a landing spot to a launch speed or angle.
- Derive results in symbols and check them with limiting cases.

Key ideas

- Working backward: write the x and y equations with the unknown in them, then use one to eliminate t.
- Check a derived result with limiting cases: set an angle to 0° or 90°, or a speed very large, and ask whether the answer makes physical sense.

1. Derive. A ball is launched at speed v_i and angle θ on level ground. Show that the time of flight is $t = \dfrac{2v_i \sin\theta}{g}$ and the maximum height is $y_{\max} = \dfrac{v_i^2 \sin^2\theta}{2g}$.

2. Derive. A ball thrown horizontally at speed v_i from the top of a cliff lands a horizontal distance d from the base. Find the cliff height h in terms of v_i, d, and g.

3. Limiting cases. Check the expressions you derived in problems 1 and 2.

a) What do $t = \dfrac{2v_i \sin\theta}{g}$ and $y_{\max} = \dfrac{v_i^2 \sin^2\theta}{2g}$ predict as $\theta \to 0$? Does that make physical sense?

b) What do they predict at $\theta = 90^\circ$? Compare with a ball thrown straight up at v_i. What is the range then?

c) In $h = \dfrac{gd^2}{2v_i^2}$, what happens to h as v_i becomes very large while d stays fixed? Does that make sense?

4. A ball launched at 20 m/s must land on level ground 30 m away. Find both launch angles that work.

5. Free throw challenge. Mr. Vincenzo's free throw leaves his hands 2.0 m above the floor at 50°. The rim is 3.05 m high and 6.0 m away horizontally. What launch speed sinks the shot?

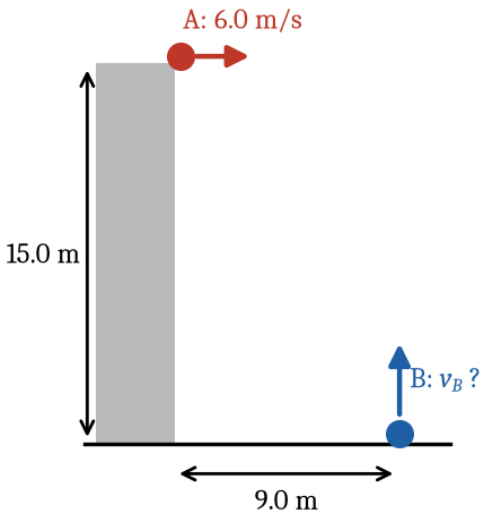
6. Two projectiles meet. Ball A is thrown horizontally at 6.0 m/s from the top of a 15.0 m tower. At the same instant, ball B is thrown straight up from the ground 9.0 m from the base of the tower.

a) Use A's x column to find when A is directly above B's launch spot.

b) Set the two vertical positions equal at that time. Show that the $\frac{1}{2}a_y t^2$ terms cancel, and find B's launch speed so the balls collide.

c) How high above the ground do they meet? Is B moving up or down then? Has A already landed?

d) Show in symbols that $v_B = \dfrac{H v_A}{d}$.



7. Paragraph argument. A classmate claims that on the Moon ($g = 1.6\ \text{m/s}^2$), ball B in problem 6 would need a different launch speed to hit ball A. Do you agree? Write a paragraph argument: a claim, evidence (a physics principle or equation), and reasoning that links them, in 3–5 complete sentences.

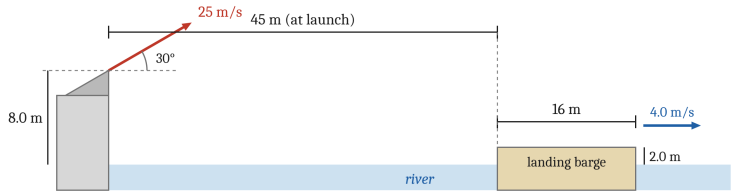
Part 6 · Putting It All Together

Tier: Extension **When:** End of the unit

About this page

- Two long, open-ended problems that use every idea in the unit: vectors, relative velocity, launches from a height, and motion graphs. Each one takes two pages.

1. Mr. Lehman's River Jump. Mr. Lehman is jumping his motorcycle across a river for a charity show. The ramp edge is 8.0 m above the water and launches him at 25 m/s, 30° above the horizontal. He must land on a barge whose deck is 2.0 m above the water and 16 m long. At the instant he leaves the ramp, the near edge of the deck is 45 m from the ramp edge, and the barge is moving straight away from the ramp at a steady 4.0 m/s relative to the shore. Use down as negative and put the water at $y = 0$.



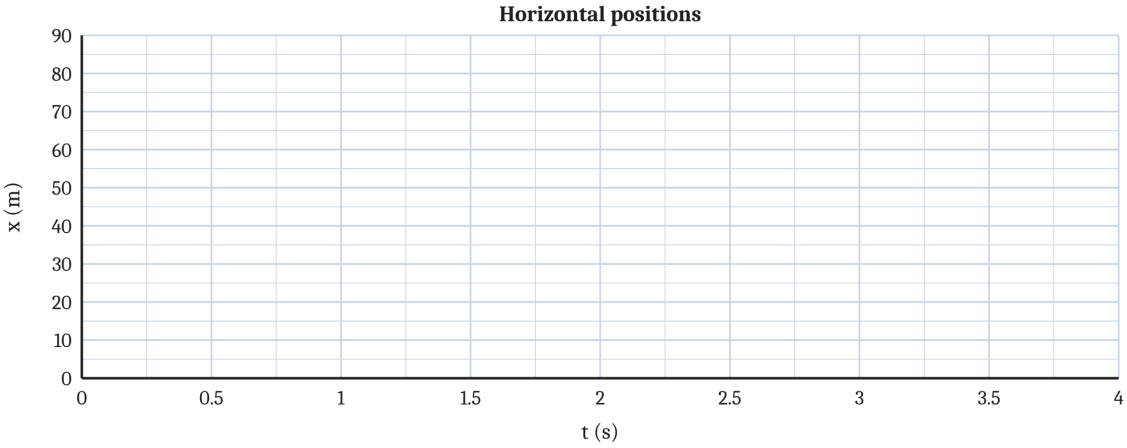
a) **Steering the barge.** The river's current flows at 1.5 m/s along the banks, sideways to the jump. To move straight away from the ramp at 4.0 m/s relative to the shore, which way must the barge point, and how fast must it move through the water? Sketch the vectors.

b) Does Mr. Lehman land on the deck? Support your answer with calculations, and say where on the deck he touches down.

c) Would the stunt have worked if the barge had stayed still? Explain.

d) How fast is Mr. Lehman moving, and in what direction, relative to the barge as he touches down?

Graph the jump. Put the ramp edge at $x = 0$. Use 4.0 m/s for the barge on every graph.



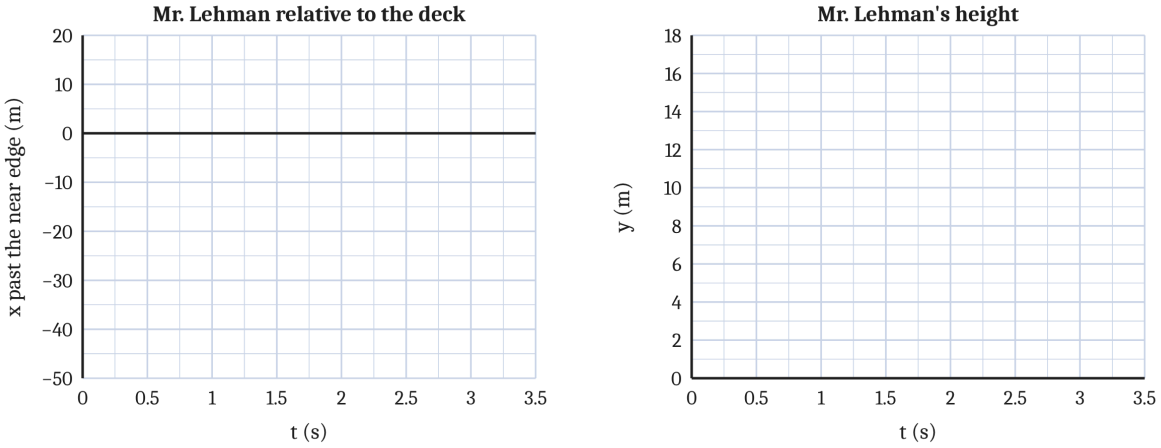
e) Graph Mr. Lehman's horizontal position and the positions of the deck's near edge and far edge from $t = 0$ to $t = 4.0$ s. Draw his line solid until touchdown, then dashed, as if he kept flying. Circle his touchdown point.

f) Find where Mr. Lehman's line crosses each barge line, first from your graph, then by setting his position equation equal to each edge's position equation. What is happening at each crossing?

g) Use the two crossing times and his touchdown time to explain why he lands on the deck. What would a touchdown before the first crossing mean? After the second?

h) What does the slope of each line stand for? Why are the two barge lines parallel, and what would they look like if the barge stayed still?

The jump as seen from the barge.

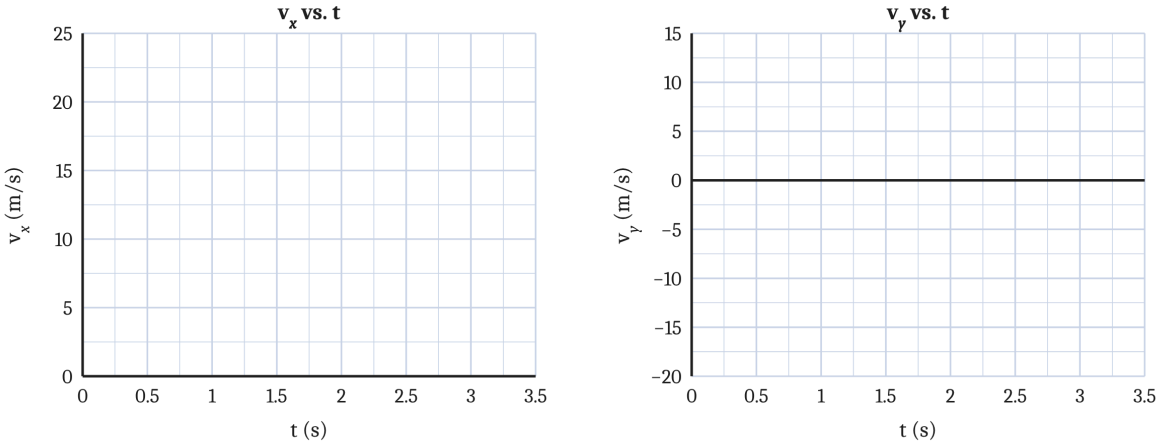


i) On the left grid, graph Mr. Lehman's horizontal position measured from the deck's near edge (positive past the edge), from launch until touchdown. Where does your line cross zero, and how does that match part f? What range of values at touchdown means he lands on the deck?

j) Why is this line straight, and what does its slope stand for? Compare it with your answer to part d.

k) On the right grid, graph Mr. Lehman's height above the water and draw the deck height as a dashed line. Mark the top of the jump and the touchdown. How does the graph show his time in the air?

Velocity graphs.



l) Graph Mr. Lehman's v_x and v_y from launch until touchdown.

m) Use the area between each line and the time axis to find Δx and Δy for the jump. The v_y graph has area above and below the axis: what does each part stand for, and why does the total match the drop from the ramp edge to the deck?

n) The producer wants him to touch down in the middle of the deck. Recommend one change to the setup, calculate the new value, and describe how your horizontal-position graph would change.

o) **Challenge.** What range of barge speeds lets Mr. Lehman land on the deck? Show the fastest and slowest cases as extra lines on your horizontal-position graph.

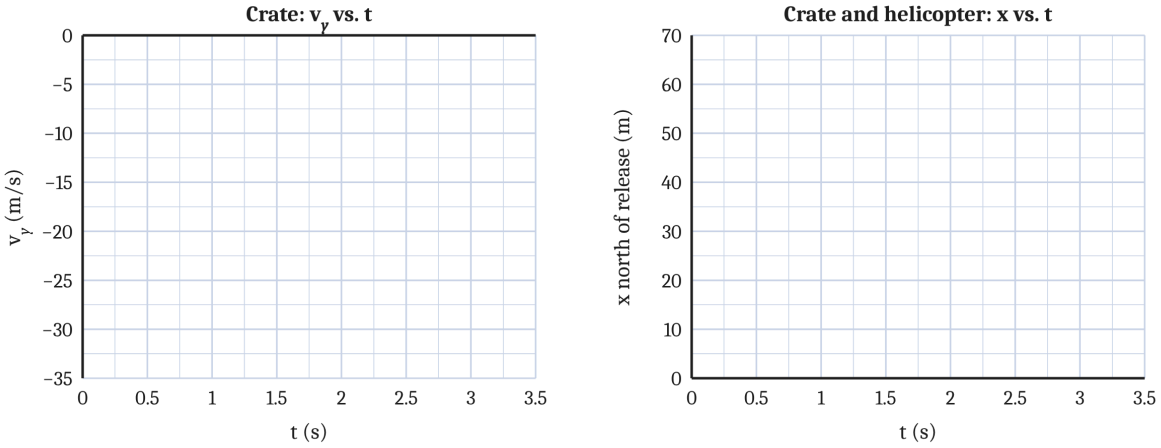
2. Mountain Rescue. Dr. Shafer tried to climb a mountain and got stuck. Mr. Jantz is flying a rescue helicopter to Dr. Shafer's camp, 30 km due north of the base. The helicopter's airspeed is 50 m/s, and a 12 m/s wind blows from west to east the whole time.

a) Which way must Mr. Jantz point the helicopter to travel due north, and how many minutes does the trip take? Sketch the vectors.

b) Near the camp he flies due north at a ground speed of 20 m/s, 45 m above the ground, and releases a supply crate. Where must he release it so it lands at the camp, and how fast is the crate moving when it lands?

c) A gust carries the crate into the trees. Dr. Shafer reaches it by following a trail: 30 m at 20° E of N, then 25 m at 60° E of N. How far, and in what direction, should Dr. Shafer walk to get straight back to camp?

Graph the crate drop. Start the clock when the crate is released.

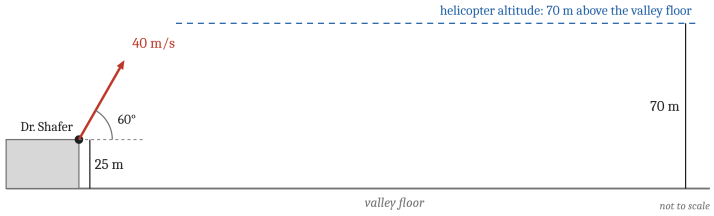


d) On the left grid, graph the crate's v_y from release to landing. Use the area between your line and the time axis to check the 45 m drop.

e) On the right grid, graph the horizontal position of the crate and of the helicopter (still flying north at 20 m/s) from release until the crate lands. What do your two lines show about where the crate is, compared with the helicopter, while it falls?

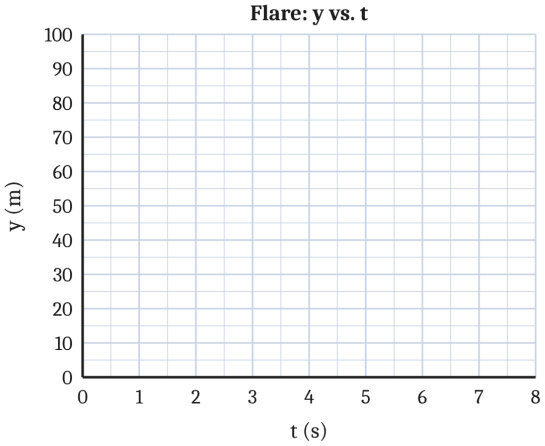
f) A real crate feels air resistance. Sketch how the crate's line on the right grid would change. Should Mr. Jantz release it earlier or later than in part b? Explain using your graph.

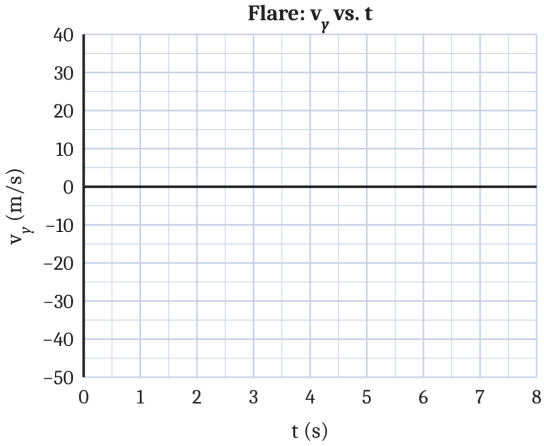
The signal flare. That night Dr. Shafer stands at the edge of a 25 m cliff and fires a flare at 40 m/s, 60° above the horizontal. Mr. Jantz hovers 70 m above the valley floor. Use the valley floor as $y = 0$.



g) Does the flare rise high enough to pass Mr. Jantz's altitude? If so, for how long is it above that altitude, and where does it land?

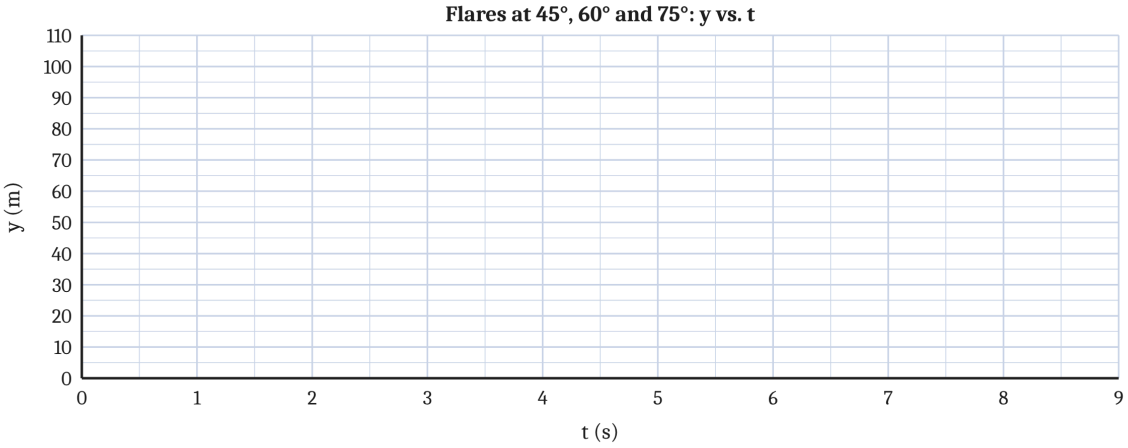
h) On the left grid, graph the flare's height from launch to landing (plot points or copy your Desmos graph). Draw a line at 70 m and mark the crossings. Draw a tangent line at each crossing and find its slope. What do the slopes stand for, and how do they compare?



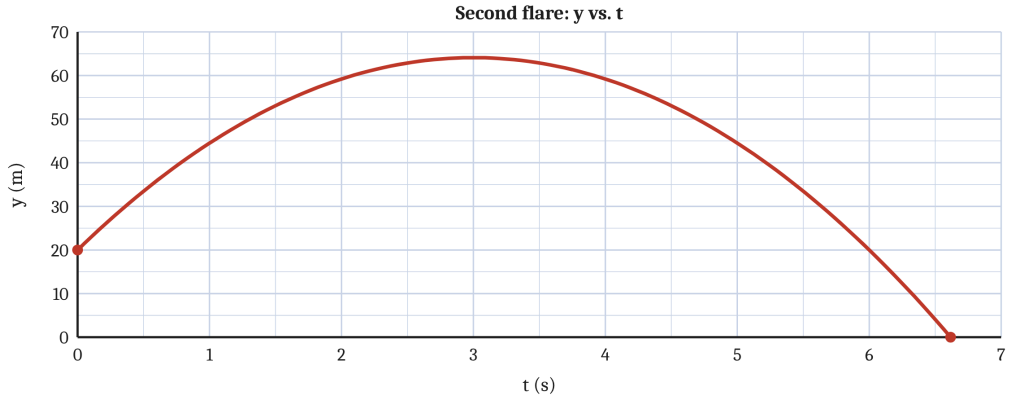


i) On the right grid, graph the flare's v_y from launch to landing. Shade the area between your line and the time axis from the first crossing of 70 m to the second. What is the total of that area, and why?

j) Dr. Shafer could fire at 45° or 75° instead. Sketch the 45°, 60° and 75° flares on the grid (find each one's v_{iy}, highest point and landing time first). Using your graph, decide which angle keeps the flare above 70 m the longest and whether every angle reaches 70 m. What feature of the graph did you use?



Reading a graph. The next night Dr. Shafer fires a second flare at 40 m/s from a different ledge. Mr. Jantz's tracker records the graph below. Use only the graph and the launch speed.



k) How high is the ledge? Find v_{iy} two ways: from the time it takes to reach the top, and from the height the flare gains.

l) At what angle above the horizontal was the flare fired?

m) About how far from the base of the ledge does the flare land?